Paired Sign Guards and a Square-Anchored Factorial Radius

Direct Diophantine Consistency Construction

24 October 2026

1 A different route: a direct quartic in 489 integers

The direct C222 proof-certificate branch admits a new ordinary polynomial

$$P_{489}(z_1,\dots,z_{489}) = \sum_{j=1}^{308} R_j(z)^2, \qquad z \in \mathbb{Z}^{489},$$

where the accompanying equation file lists every integral quadratic R_j. Its expanded polynomial has exact degree 4, 6905 nonzero monomials, and coefficient height 11302118. A simpler sibling has 493 integer variables, 310 residuals, 7174 monomials and height 10945213. Both improve the inherited 517-integer direct quartic, not the much-higher-degree eleven-variable frontier.

These are fully materialized polynomials, with no hidden natural variables, variable exponents or unspecified program constants. Their interpretation as consistency equations inherits the existing ordinary C222/ZFC and signed-Pell calibration. Subject to precisely that calibration, the new reductions prove

$$\mathrm{ZFC} \vdash \big(\forall z \in \mathbb{Z}^{489}\ P_{489}(z) \neq 0\big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

No new kernel-checked internal derivation, proof of consistency, global minimality, or historical priority is asserted. The retained source closure includes the finite calculus and its pinned calibration, not just a compiler acting on an unnamed predicate.

Attribution and the different measurement stages. The paired-product principle already appears in the upstream six-anchored- guard construction at commit `59ee8f2` (24 October 2026, 22:32 UTC). Its full accepted review reports 56 predicates, 1436 coordinates and 1242 rows in the *unprojected* parent, saving 18 coordinates and six rows. It makes no projected-count claim. Its six pairs use retained scalar anchors before power recovery. Here twelve pairs are restored after power recovery on the later factorial-carrier source; the measured projections are $517 \to 493$ and, with the further F^2 change, 489. We credit the overlapping product-sign principle and claim no priority for it. The authenticated late review is preserved with these artifacts.

1.1 Two signs need one guard when their sum is known

For integers r, s with $r + s \geq 0$,

$$r, s \geq 0 \quad \Longleftrightarrow \quad \exists a, b, c \in \mathbb{Z}: \quad rs = a^2 + b^2 + c(c+1). \tag{PG}$$

Indeed the right-hand side is nonnegative, so a negative r would force $s \leq 0$, contradicting the sum hypothesis; interchange r, s for the other case. Conversely every natural n is $a^2+b^2+c(c+1)$: represent $4n+1$ by three squares, permute its unique odd coordinate to $2c+1$, and divide the other coordinates

by two. This is the usual consequence of Legendre's three-square theorem. The sum hypothesis cannot be omitted: $r = s = -1$ has positive product.

The direct source has twelve digit remainder/slack pairs: two exact-binomial pairs and ten range-binomial pairs. Each has an existing row

$$r + s = q^k - 1,$$

where k is one of the positive indices $t, t+1, y+1, x+1$. Replacing their twenty-four separate three-witness guards by twelve copies of (PG) preserves quadratic residuals. After all monic eliminations, this reduces 517 variables to 493, rather than merely subtracting thirty-six from an unreduced variable count.

1.2 The signs are recovered after, not before, the powers

Keep the factorial ceiling remainder and slack f_r, f_s, and the first exact-digit output slack o_0, separately nonnegative. Their raw rows give

$$D = f_r + f_s + 1 \geq 1, \qquad q = D + o_0 + 1 \geq 2. \tag{B1}$$

All seed-exponent guards and encoded-representative guards are retained. Together with the shared square budget, they establish the signed indexed-Pell hypotheses independently of the twelve removed digit signs. The request q^t orients the t-indexed pair; thus $V = (\rho+1)^t \geq 1$ has the correct positive value. The unchanged ceiling identity is

$$DF = V + f_s, \tag{B2}$$

which forces $F \geq 1$ and then $V \geq 2$.

The retained guard $B \geq 0$, with $C + F^2 + 1 = Bi_q$ and $C \geq 1$, gives $B > 0$. The signed request B^K gives its positive output N, except possibly for the inverse case $B = 1, N = -1$. In either case the other K-indexed base $N + V + 1$ is at least two. It orients the shared pair, excludes the inverse case, and recovers $N = B^K > 0$ and $q = N(N+V+1)^K$. Every remaining index has a request at base q or $q+1$. All nineteen powers are therefore recovered before any replaced sign is used. Now $q^k - 1 \geq 0$, so (PG) restores each original remainder and slack sign. The old carry-free digit, amplified-CRT and Horn-flow proofs apply.

In the reverse direction an old semantic certificate has $r, s \geq 0$, so each product rs receives three witnesses by (PG). This is an existence-preserving map, not an identification of old and new guard coordinates.

1.3 An existing square can guard the factorial radius

The stronger endpoint changes only three source rows:

$$\rho = F^2 t^2, \qquad y = F^2 + r_0 + 1, \qquad R = 2 + t + x + y + F^2, \qquad K = 2R. \tag{SQ}$$

The old rows used F in place of F^2 at these three positions. The wire F^2 already exists for the reciprocal product $B = \binom{F^2+t}{t}$. No extra square coordinate is introduced; instead, the independent nonnegativity guard on F is removed.

Before interpreting F, its square gives $\rho \geq 0$, $y \geq 1$ and $R \geq 19$, since $t = t_0 + 16$ and $x, r_0, t_0 \geq 0$. Thus all seed indices remain positive: $t, K, \rho + t, F^2 + t, x+1, y+1$, the positive encoded representatives and the positive derived label sum. The new ordered lower-bound certificate establishes these facts directly from the modified polynomial; it does not reuse an assumption $F \geq 0$. Equations (B1)–(B2)

and the preceding power recovery now force $F \geq 1$. Only after this step is F used as a natural source parameter.

Once $F \geq 1$, the larger radius satisfies $\rho \geq Ft^2$. The new R dominates the old $2 + t + x + y + F$, so the same local majorant exponent remains valid. When the factorial is recovered, $t \geq 16$ gives $F = t! \geq 16! > 2^{44}$; hence the old bounds

$$L, J < R^E < 2^R < 2^K, \qquad 3\ell(t+1) \leq 2F \leq K$$

continue to validate the source and derived-label range. The literal compiler then removes the factorial guard and one additional monic wire, giving 489 variables.

1.4 A self-bounding factorial at any sufficiently large radius

Let t, F, ρ be positive integers, $\rho \geq Ft^2$, and put

$$V = (\rho+1)^t, \qquad D = \binom{\rho+t}{t}.$$

There are natural f_r, f_s satisfying

$$D = f_r + f_s + 1, \qquad DF = V + f_s \tag{F}$$

if and only if $F = t!$. To prove this write $L = t!$. The exact product identity gives

$$\frac{V}{D} = L \prod_{i=1}^{t} \left(1 - \frac{i-1}{\rho+i}\right), \qquad L\left(1 - \frac{1}{2F}\right) < \frac{V}{D} \leq L.$$

The lower bound follows from $\sum (i-1)/(\rho+i) \leq t(t-1)/(2\rho) < 1/(2F)$. Rows (F) say $F - 1 < V/D \leq F$, so $F \leq L$. If $F < L$, then $L \geq F+1$ and $L(2F-1) \geq 2F^2$; consequently $L - L/(2F) \geq F$, contradicting the strict lower bound. Thus $F = L$. Conversely, for $F = L$, the ratio lies in $(L-1, L]$; choose $f_s = DL - V$ and $f_r = D - f_s - 1$.

The polynomial does not assume the proposed D is already this binomial. Recover powers first. The common radix satisfies $q > V^2$. Once (B2) has forced $F \geq 1$, one has $\rho \geq t$, and every coefficient through degree t in $(q+1)^{\rho+t}$ is at most $\binom{\rho+t}{t} \leq V < q$. Thus the first exact digit is valid and (F) forces $F = t!$. For the second exact binomial use $F^2 + t \leq (\rho+1)^2$, giving coefficient bound $V^2 < q$. Only then identify B, derive $B \geq 2$, and validate the remaining ranges. This order avoids a circular interpretation of the product binomial.

1.5 Completeness and the exact polynomial maps

The square-anchored variant is reconstructed from a finite proof, not obtained by keeping the old tuple unchanged. Choose the padded horizon $t \geq 16$ beyond the finite proof-tree length, local parameters and label sums, then set $F = t!$. Construct the positive Horn-flow accumulators with weights $1 + jF$. Choose a multiple y of $t!$ exceeding every accumulator and F^2, so $r_0 = y - F^2 - 1 \geq 0$. The beta CRT gives x and the endpoint quotients. Define ρ, R, K by (SQ), construct the reciprocal CRT representatives at precision $2K$, and then define the powers and radix. The wider-radius lemma supplies the factorial ceilings; the original carry bounds supply all digit witnesses; simultaneous Pell completeness supplies the shared frame. Finally apply (PG) to every product.

The Horn factorization is an exact polynomial identity. Every subsequent coordinate deletion has an integral monic restoration; reversing the recorded list supplies the inverse of projection. The final JSON contains all 308 residuals and all 489 coordinates, and the independent expanded coefficient list verifies degree, support and height. The full proof, deterministic builder, lower-bound certificates, changes to the three radius rows, and both projection/restoration maps accompany this section. Finite regression tests supplement these universal arguments; they do not fabricate a zero or test consistency.